

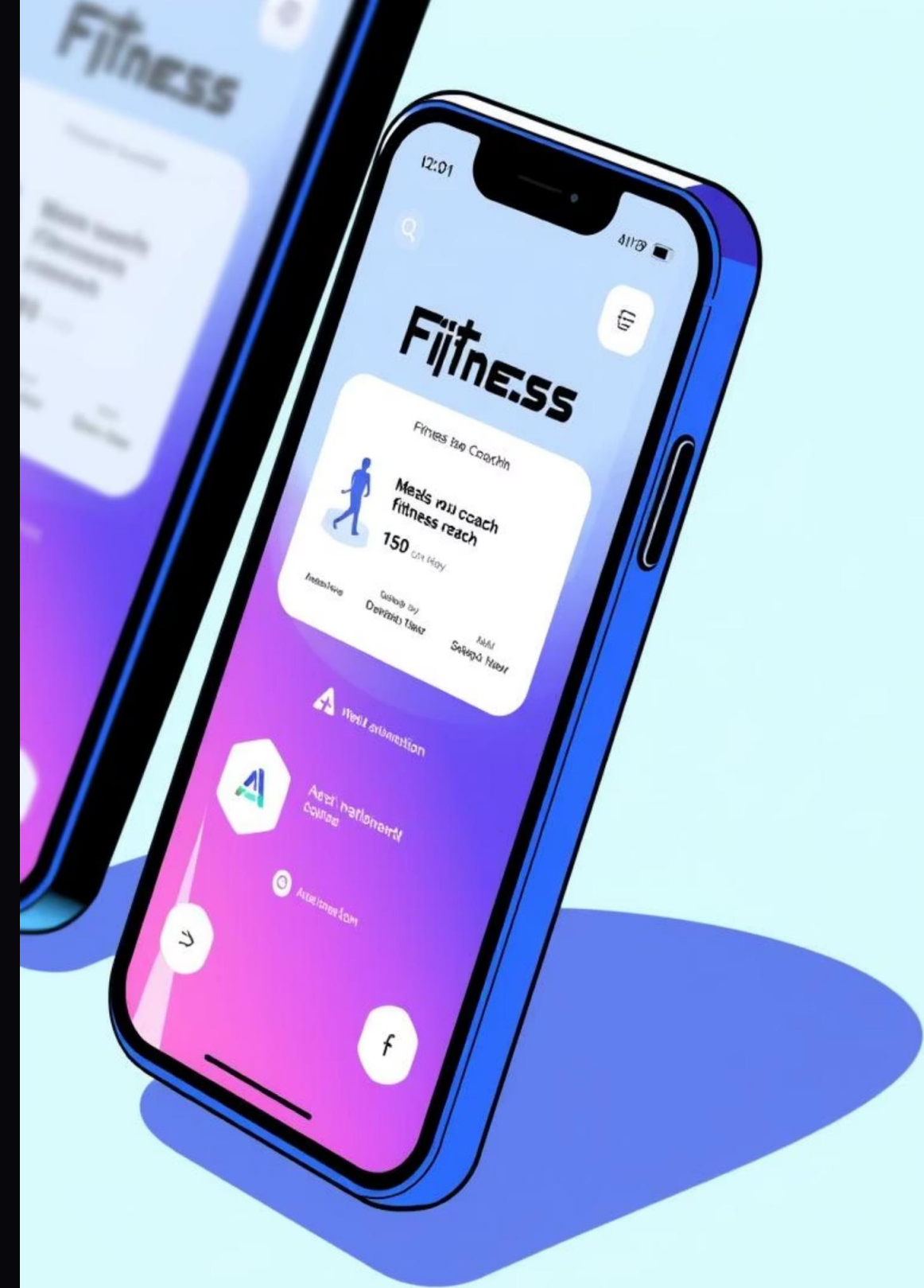
Fitness Coach Mobile Application

Final Year Project Presentation

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Introduction

→ Growing Demand

The fitness industry is witnessing a significant surge in demand for personalised guidance, particularly through convenient mobile technology.

→ Empowering Wellness

Our project focuses on developing an intuitive fitness coach application, designed to empower users on their wellness journeys with comprehensive support and insights.

→ Accessible Tools

Mobile applications have emerged as highly accessible and effective tools for individuals seeking to improve their health and receive professional coaching.



Problem Statement

Lack of Personalised Coaching

Many individuals struggle to access affordable, personalised fitness coaching that is available at their convenience, anytime and anywhere.

Limited App Integration

Existing fitness applications often fall short in providing seamless integration of real-time progress tracking with truly customised workout and nutrition plans.

Motivation & Adherence Challenges

Users frequently face significant hurdles in maintaining motivation and adherence to fitness routines without interactive support and consistent engagement.



Objectives

1

User-Friendly Design

To design an intuitive and highly user-friendly mobile application that offers personalised fitness coaching.

2

Core Features Implementation

To implement essential features including robust workout planning, accurate progress tracking, and constructive motivational feedback mechanisms.

3

Scalability & Adaptability

To ensure the application is scalable and adaptable, catering to a diverse range of user fitness levels and varying personal goals.

4

Coach Management Platform

To provide a comprehensive platform enabling fitness coaches to remotely manage and effectively engage with their clients.



Technologies Used



Frontend: Flutter

Utilised for cross-platform mobile development, ensuring a consistent user experience across both iOS and Android devices with a single codebase.



Backend: Dart

Powers the robust API services, providing efficient data handling and communication between the mobile client and the database.



HuggingFace API's

Leveraged for hosting and providing a scalable infrastructure, ensuring high availability and performance for the application.



Additional: Firebase

Integrated for real-time notifications, enhancing user engagement, and providing secure authentication services.

System Architecture



Key Features

- Personalised Workout Plans

Generate custom workout routines tailored to individual user goals and available equipment.

- Progress Tracking

Monitor key metrics such as workouts completed, calories burned, and body statistics over time.

- Progress Photo Review

Ability to upload and review progress photos for visual motivation and tracking transformation.

- Session Scheduling

An integrated tool for conveniently booking and managing coaching sessions with ease.



Testing & Validation



Functional Testing

Thorough testing of all application modules across Android and iOS devices to ensure intended functionality.



Usability Testing

Engaging with target users to gather feedback and refine the user interface and overall user experience flow.



Performance Testing

Evaluating the app's responsiveness and stability under various load conditions to guarantee smooth operation.



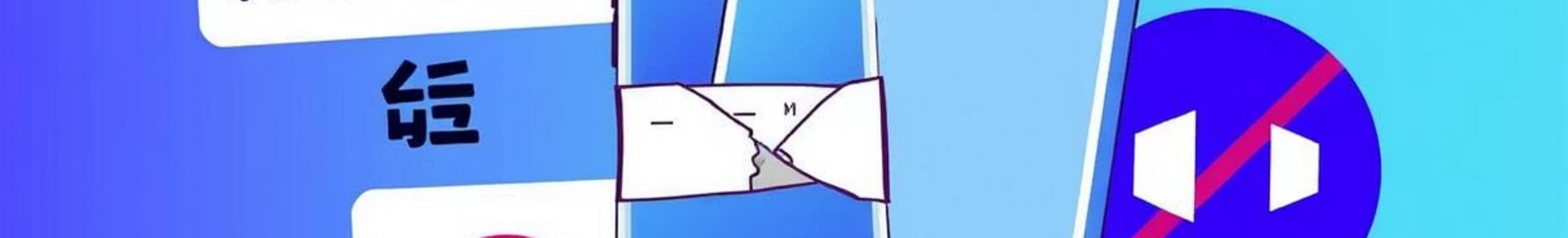
Integration Testing

Rigorous checks to ensure robust data protection, privacy compliance, and safeguard user information.



Unit Testing

Incorporating valuable feedback from a pilot group of fitness enthusiasts to drive continuous improvements.



Limitations



Language Support

The initial version is currently limited to English language support only.



Sensor & API Reliance

Dependence on the accuracy of user device sensors and third-party health APIs for data collection.



Offline Functionality

Offline capabilities are restricted, requiring an internet connection for full access to all features.



Interaction Scope

Coach-client interaction is presently limited to in-app messaging, with no live video call functionality.

Future Work



Multi-language Support

Expand accessibility by implementing support for multiple languages.



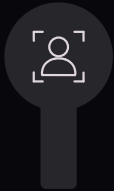
AI-Driven Recommendations

Integrate AI for adaptive workout recommendations based on individual user progress and performance.



Live Coaching & Groups

Introduce live video coaching sessions and the option for group workout activities.



Enhanced Offline Capabilities

Improve offline functionality through advanced local data caching mechanisms.



Advanced Coach Analytics

Develop a more comprehensive analytics dashboard for coaches, offering deeper insights into client progress.

Our vision is to evolve the Fitness Coach Mobile Application into a leading, intelligent, and fully integrated wellness platform.